Saransh Agrawal

saranshagarwal2020@gmail.com | linkedin.com/in/saransh-agarwal8899 | Website | GitHub

Education

Texas A&M University — **Aug 2023 – May 2025**
Master of Science in Data Science: GPA: 3.7/4.0 — *College Station, TX*
- Research Assistant at FLAIR Lab - Responsible NLP, Safety, Robustness.
- Teaching Assistant, Dept. of Statistics (Mentored 50+ students)

Manipal Institute of Technology — **July 2017 – Aug 2021**
Bachelor of Technology in Control Systems Engineering — *Manipal, IN*

Technical Skills

Languages: Python, Go, C++, SQL.
ML/AI Frameworks: PyTorch, Transformers, TensorFlow, Scikit-Learn, OpenAI, HuggingFace, vLLM.
MLOps & Infrastructure: Docker, Kubernetes, Apache Kafka, AWS (EC2, SageMaker, S3), SLURM, Jenkins CI/CD, Git.
Data & Storage: PostgreSQL, MongoDB, Elasticsearch, Apache Spark, Pandas, NumPy.
Areas of Expertise: Applied Machine Learning, Large Language Models, Natural Language Processing (NLP), Reinforcement Learning, Deep Learning, Distributed Systems.

Publications

Saransh Agrawal and Kuan-Hao Huang. Selective Amnesia – Constrained Unlearning for Large Language Models via Knowledge Isolation. In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025) at ACL.*

Projects

- **VLM-LongContext: Efficient Long-Context Reasoning in Multimodal LLMs** — *Python, PyTorch, LLaVA*
  - Built ML **optimization** framework reducing memory consumption by **75%** for long-context multimodal reasoning using Streaming LLM and Cache Merging.
  - Implemented image chunking with descriptive generation enabling processing of 100+ images vs 6-8 baseline limit on MileBench dataset.
  - Benchmarked LLaVA-1.5, MiniGPT-4, and LLaVA-OneVision models, achieving optimal memory-performance tradeoffs.
- **Preference Vector Framework for LLM Alignment** — *Python, PyTorch, Transformers, DPO*
  - Collaborated with National Taiwan University (5-person international team) to design modular **DPO** framework.
  - Enabled **controllable** post-training alignment for model **personalization** by dynamically merging preference vectors at test time with user-adjustable scaling coefficients.
  - **Outperformed** Safe-RLHF baselines with **4x faster** training. ***ACL ARR*** (***preprint***).
- **Satellite-based Crop Monitoring System (SCMS)** — *Python, FastAPI, Docker, React.js, MongoDB*
  - Led 4-person team building data ingestion, transformation, and serving pipelines for satellite imagery processing.
  - Integrated PlanetScope satellite API for automated imagery acquisition with cloud filtering and AOI-specific preprocessing.
  - Deployed **LSTM-based time-series forecasting** models to predict outcomes based on sequential satellite data, achieving an R^2 **of 0.81**. *Presented at ASA Conference 2024.*
- **Graph-Based Ranking & Semantic Search Engine** — *Go, Python, Apache Arrow, Sentence-Transformers*
  - Engineered an end-to-end **recommendation** system for 70k+ ACL Anthology papers, improving discovery of the most relevant and influential content.
  - Constructed a large-scale citation graph and implemented the PageRank graph algorithm from scratch to compute paper authority scores for **targeted content discovery**.
  - Built hybrid search system combining graph-based metrics and **sentence embeddings** to model complex relationships.

Work Experience

Software Developer, ML Team — **July 2021 – July 2023**
Viewzen Labs — *Bangalore, India*

- Led a high-stakes predictive modeling initiative to identify at-risk individuals from a **highly imbalanced** dataset of 45,000+ women with over 300 features.
- **Architected** and spearheaded the development of a **scalable**, distributed AutoML platform (**MLaaS**) designed for **high-throughput**, processing over 800K requests monthly for 12+ enterprise clients.
- Designed a **robust** microservices architecture using **REST APIs** and **Kafka** Streams to handle **millions** of data points daily, ensuring system reliability and maintainability.
- Built Python library implementing AutoML logic using Hyperopt, Scikit-Learn, reducing ML development time by 80%.
- Created RESTful Python microservices with Docker containerization. Mentored 2 junior developers on API design.
- Enhanced processing speeds by 20% through C++ optimization and **concurrency** for **high-volume data transformation**.

Machine Learning Engineer Intern — **Feb 2021 – June 2021**
Centre for Digital Financial Inclusion — *Delhi, India*

- Transformed **raw audio signal** to its **MFCC** components using custom TensorFlow layers.
- Implemented RNN, GRU and LSTM architectures for text prediction at character, word and phoneme level.
- Utilized **DeepSpeech** transfer learning on Indian Accent dataset achieving **0.10 CER** and **0.17 WER** with KenLM language model. Added **T5 Transformer** to generate **SQL queries**.